

Harjeet Singh Chahal

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Education

Rutgers University - New Brunswick, New Jersey, USA M.S. in Computer Science • Relevant Coursework: Machine Learning, Massive Data Mining, Interpretable & Explainable AI, Database Systems for Data Science, Data Structures & Algorithms, Artificial Intelligence, Visual Computing.	Aug 2025 - May 2027 GPA: 4.00/4.00
Netaji Subhas University of Technology - New Delhi, India B.Tech. in Computer Science & Engineering (Artificial Intelligence) • Relevant Coursework: Deep Learning, Natural Language Processing, High Performance Computing, Distributed Computing, Operating Systems, Data Communication, Game Theory, Design and Analysis of Algorithms, Web Technology.	Nov 2020 - Jun 2024 GPA: 3.74/4.00

Technical Skills

Languages: Java, Python, C++, Go, SQL, JavaScript, R, HTML/CSS
ML / AI: PyTorch, Scikit-Learn, Hugging Face Transformers, NLTK, LangChain, RAG, OpenCV, NumPy, Pandas
Backend / Frontend: FastAPI, Flask, Node.js, REST, GraphQL, gRPC, MCP, React, React Native, Next.js, Expo
Cloud & DevOps: AWS, GCP, Docker, Kubernetes, Linux, Git, CI/CD, SLURM, Prometheus, Grafana
Data Tools: PostgreSQL, pgvector, MySQL, MongoDB, Redis, FAISS, Pinecone, ChromaDB, Kafka

Experience

Rutgers School of Health Professions Research Assistant : Mobile + ML Engineering • Productionized HIPAA-compliant PyTorch EfficientNet-B0 on FastAPI, cutting CPU inference to 25–40 ms . • Deployed Dockerized FastAPI ML APIs on GCP Cloud Run with CI/CD for automated tests, builds, and releases. • Built an Expo/React Native iOS/Android app with JWT auth, reusable UI, deep linking, persistent state, and weather, geolocation, and UV-index APIs; added a context-aware educational chatbot and engagement workflows. • Built a 50-test Python/FastAPI suite for ML inference, auth, edge cases, and parity; resolved ARM/x86 floating-point drift and regressions and owned 10+ Apple TestFlight releases tested by 100+ participants .	Piscataway, NJ Apr 2026 - Present
Technical Consulting and Research Inc. Software Development Intern : Backend + AI Engineering • Architected a local-first compliance platform (FastAPI, React, PostgreSQL/pgvector) running LLM inference and embeddings on-premise via Ollama , keeping client evidence off third-party APIs. • Built a pipeline restricting citations to retrieved chunk IDs, with server-side source resolution to reduce unsupported claims. • Developed a MCP server exposing 20 read-only compliance tools for agentic retrieval, evidence lookup and control analysis.	Weston, CT Jun 2026 - Aug 2026
Dhruv Research Associate : Analytics + Data Engineering • Engineered Python/SQL ETL pipelines with Pandas/NumPy to clean, validate, and normalize 200K+ survey responses across 10 state projects, automating QA and transformations to reduce turnaround time by 30% . • Predicted vote share across 500+ constituencies using Python/SQL, statistical weighting, and MAE/RMSE validation. • Worked within and coordinated an 8-member team , using Python/SQL trackers for sampling and demographic skew.	Gurugram, India Sep 2024 – May 2025

Projects

ScaleTrain : HPC Training Pipeline PyTorch, DDP, SLURM, NVIDIA A100 GitHub  • Built fault-tolerant PyTorch DDP training on SLURM, scaling a 30M-param transformer to 4 A100s at 500K tokens/s . • Engineered signal-driven checkpoints and deterministic data partitioning for exact resume with zero lost work . • Automated SLURM setup, fault injection, benchmarking, metrics, and reporting in an idempotent pipeline with 64 tests .	May 2026 - July 2026
VideoQuery : Semantic Video Search Whisper, Celery, Redis, AWS GitHub  • Built multimodal video search and grounded Q&A with Whisper, CLIP, BLIP, and pgvector using timestamped citations. • Fused BM25, dense, CLIP, and caption retrieval with RRF + reranking, achieving 88.5% R@1 and 3.9s temporal error. • Built crash-safe, resumable video ingestion at 10× real time; deployed on AWS EC2, S3, ECR and CloudFront.	Oct 2024 - Dec 2024
AgentXAI : Traceable Agent Explainability LangChain, FAISS, FastAPI, Streamlit Report  • Addressed unfaithful explanations in multi-agent AI , where post-hoc rationales can diverge from the actual reasoning path. • Designed a 5-agent architecture with typed traces, causal DAGs, evidence constraints, and counterfactual ablations. • Achieved 93% explanation sufficiency and 0.71 stability , producing auditable, execution-grounded rationales.	Jan 2026 – May 2026

Achievements

Programming: Solved 1,550+ LeetCode problems with a 1,300+ day active streak. | leetcode.com/u/harjeetchahal
Academic Distinction: Ranked top 0.15% nationally among ~1M candidates in JEE Main 2020; GRE Quant: 166/170.